

Computational Notes on Minimal Goldbach Witnesses, Q-free Wall-Spectrum Triage, and Q-line Interval Coverage

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Abstract

For an even integer N , define the minimal Goldbach witness

$$p(N) = \min\{p : p \text{ is prime and } N - p \text{ is prime}\}.$$

This note reports finite computational experiments around three related objects: minimal Goldbach witnesses, q-free pre-witness triage for unusually large witnesses, and fast interval coverage. It is not a proof of the Strong Goldbach conjecture, not a certified extension of the known verification up to $4 \cdot 10^{18}$, and not a claim that q-line interval coverage is new. The q-line coverage used here belongs to the classical interval-cover lineage of Shen and of Deshouillers, te Riele, and Saouter (Refs. 1 and 4).

The present repository contributes a native C/C++ implementation, compact data, and reproducible finite measurements under explicitly stated primality tests. The minimal-witness table covers

$$N_e = 2 \cdot 10^e, \quad e = 2, \dots, 5000,$$

with 4999 probable-prime computational witnesses under the GMP primality test used here. In this table, the largest absolute witness is

$$p(2 \cdot 10^{4718}) = 549667,$$

and the largest normalized witness is

$$\frac{p(2 \cdot 10^{3903})}{\log(2 \cdot 10^{3903})} = 58.300567.$$

A later sparse offset experiment at

$$N = 2 \cdot 10^{15000} - 77368$$

found

$$p(N) = 1,223,309, \quad p(N)/\log N = 35.417712438.$$

This row was selected by a q-free scorer and then checked by an increasing minimal-witness search to $p_{\max} = 2,000,000$. It is a probable-prime computational witness under GMP, not a formal primality certificate.

The q-free wall-spectrum benchmark asks whether features computed before the exact witness search can enrich for large-witness rows. On nine Oliveira e Silva–Herzog–Pardi least-prime holdout windows, the frozen K3 adaptive policy covers 413/450 benchmark target rows while opening about 368 rows per window, compared with 346/450 for a dense wide baseline opening about 808 rows. This is an enrichment result, not a no-skip predictor.

Finally, a separate q-line campaign verifies, under the same probable-prime convention, $1.002 \cdot 10^{12}$ consecutive even values starting at $2 \cdot 10^{19} + 2$. The q-lines covered 99.999994902% of rows and direct fallback verified the remaining 51,083 rows, with no unresolved rows. These are computational witnesses, not formal primality certificates.

Repository: <https://github.com/alexkulikovskys/goldbach-computational-lab>

1 Position of This Note

The even, or Strong, Goldbach conjecture asserts that every even integer greater than 2 is a sum of two primes. This note studies three finite computational objects related to that conjecture.

The first object is the minimal witness $p(N)$. It records how far an increasing search must go before the first decomposition

$$N = p + q$$

is found. This is a stricter measurement than merely finding some decomposition, because the first successful p is retained. Least-prime Goldbach decompositions and related record sequences are known objects; see, for example, Deshouillers, te Riele, and Saouter (Ref. 4) and the OEIS entries A025017, A025018, and A025019 (Ref. 7).

The second object is q-free wall-spectrum triage. It asks whether features computed before the exact witness search, and without testing whether $q = N - p$ is prime, can enrich for rows with unusually large $p(N)$.

The third object is interval coverage. It asks for at least one decomposition for every even N in a stated finite interval. It does not try to find the minimal witness for each row. Its purpose is speed: cover many even values at once, then directly verify any holes.

The distinction matters. A minimal-witness profile is useful for studying background, spikes, and possible predictors of hard points. Wall-spectrum triage is a tested way to choose promising neighborhoods before doing exact witness work. Q-line interval coverage is useful for quickly checking a finite block, but it does not by itself locate the rows with unusually large $p(N)$.

The work reported here should therefore be read as a reproducible computational study and implementation note. It does not supersede the known complete verification up to $4 \cdot 10^{18}$ by Oliveira e Silva, Herzog, and Pardi, and it should not be cited as a certified extension of that result (Ref. 6). The q-line campaign begins above that numerical endpoint, but the evidentiary standard is different: the present rows use GMP probable-prime tests and have not been upgraded to independent primality certificates.

2 Reproducibility Manifest

The paper is intended to be read together with the public repository. Detailed file locations, compact result snapshots, public native tools, exact parameter settings, prior-art guardrails, and current limitations are maintained in the AI/reviewer context note:

[docs/ai-context-for-paper.md](#)

This keeps the PDF readable while preserving a stable map for reproducing and auditing the computations.

The GMP workers use `mpz_probab_prime_p`. GMP returns 2 for a definitely prime value, 1 for a probably prime value, and 0 for a composite value (Ref. 9). In this note, rows accepted by the workers are therefore called probable-prime computational witnesses unless an independent certificate is explicitly mentioned.

3 Residue Classes Modulo 6

Every prime greater than 3 is congruent to either 1 or 5 modulo 6. This is only a necessary condition, but it is the correct first filter after removing multiples of 2 and 3.

For pairs in which both primes are greater than 3, the possible residue lanes are shown in Table 1.

Table 1: Allowed prime lanes for $N = p + q$ modulo 6.

$N \bmod 6$	possible $(p \bmod 6, q \bmod 6)$
0	$(1, 5)$ or $(5, 1)$
2	$(1, 1)$
4	$(5, 5)$

For the decimal scale-point sequence

$$N_e = 2 \cdot 10^e,$$

we have $10^e \equiv 4 \pmod{6}$ for all $e \geq 1$, hence

$$N_e \equiv 2 \pmod{6}.$$

Therefore, except for the small exceptional witness $p = 3$, every successful pair on this sequence must satisfy

$$p \equiv 1 \pmod{6}, \quad q \equiv 1 \pmod{6}.$$

The completed scale-point data match this exactly: among 4999 rows, 4984 witnesses are 1 mod 6, 15 are the exceptional value 3, and none are 5 mod 6.

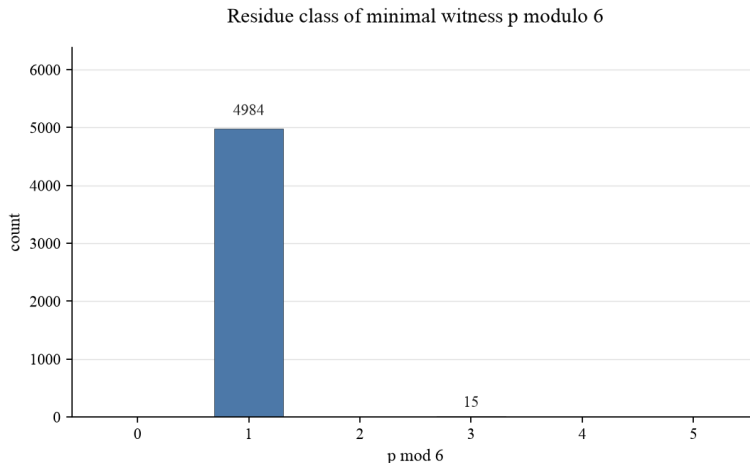


Figure 1: Residue class of $p(N_e)$ modulo 6. Non-exceptional witnesses lie in the 1 mod 6 lane.

4 Minimal-Witness Computation

For a fixed even integer N , the minimal-witness workers test prime candidates p in increasing numerical order. For each candidate they form

$$q = N - p.$$

The first accepted candidate is recorded as $p(N)$. Thus the word *minimal* always refers to the ordering of p , not to the size of q , and not to a later choice among many possible decompositions.

The search uses three filters.

1. Candidate p must be prime and below the configured bound $p_{\max}$.
2. The pair (p, q) must lie in an admissible modulo-6 lane. This removes only candidates for which q is forced to be divisible by 2 or 3, except for explicit small-prime cases.
3. The remaining q is tested for primality. In the GMP workers this is `mpz_probab_prime_p` with 25 repetitions for accepted rows.

The public scale-point table uses

$$N_e = 2 \cdot 10^e, \quad e = 2, \dots, 5000,$$

with $p_{\max} = 1,000,000$. Every reported row found a witness below this bound. The resulting rows are probable-prime computational witnesses under the primality test used by the worker.

The contiguous C verifier scans every even N in a finite interval. It uses pthread workers, a segmented bitset sieve for odd q , base primes up to the square root of the interval endpoint, and a $2 \cdot 3 \cdot 5 \cdot 7 = 210$ wheel prefilter. For $N \equiv 0 \pmod{6}$, both admissible lanes are merged in increasing p order, so the recorded value is the global first witness.

Table 2: Main parameters used in the minimal-witness runs.

Computation	parameter	value
Contiguous range runs	CPU workers	18
Contiguous range runs	segment width	5,000,000
Contiguous range runs	$p_{\max}$	100,000
Scale-point run	exponents	$2, \dots, 5000$
Scale-point run	$p_{\max}$	1,000,000
Scale-point run	GMP repetitions	25
Survivor-pressure scan	exponents	$201, \dots, 5000$
Survivor-pressure scan	<code>smallLimit</code>	250,000
Survivor-pressure scan	<code>factorLimit</code>	250,000

5 Contiguous Range Results

Table 3 summarizes two local finite-interval minimal-witness runs. Both used 18 CPU workers, segment width 5,000,000, and candidate limit $p \leq 100,000$.

Table 3: Contiguous finite-interval minimal-witness runs.

Range	even values	time	failures	mean attempts	median	q99	max p
20B...100B	40,000,000,001	54.638s	0	8.336624	6	35	2803
100B...1T	450,000,000,001	718.832s	0	9.056742	7	39	3457

The maximum-witness cases in these two runs were

$$35,884,080,836 = 2803 + 35,884,078,033$$

and

$$599,533,546,358 = 3457 + 599,533,542,901.$$

These runs are modest in mathematical scope, but useful as local performance and background measurements.

6 Scale Points Through $e = 5000$

The decimal scale-point table contains 4999 rows:

$$N_e = 2 \cdot 10^e, \quad e = 2, \dots, 5000.$$

The repository includes a compact table containing the exponent, witness p , residue class, normalized ratio, attempt counts, and verification flag. The enormous $q = N_e - p$ values are omitted from the public table to keep the repository compact. Summary statistics are shown in Table 4.

Table 4: Summary of the full scale-point table.

Statistic	value
completed scale points	4999
median $p(N)$	14407
mean $p(N)$	31973.490098
90th percentile $p(N)$	84223
95th percentile $p(N)$	121789
99th percentile $p(N)$	225157
maximum $p(N)$	549667 at $e = 4718$
median $p(N)/\log N$	3.277880
mean $p(N)/\log N$	5.289201
99th percentile $p(N)/\log N$	27.782478
maximum $p(N)/\log N$	58.300567 at $e = 3903$

The bucketed behavior by exponent range is shown in Table 5. The absolute witness tends to grow with scale, but unevenly. The normalized quantity $p(N)/\log N$ remains noisy and spike-driven rather than monotone.

Table 5: Scale-point buckets by exponent.

e -range	points	median p	p99 p	max p	median $p/\log N$
2–1000	999	2659	38977	64879	2.939861
1001–2000	1000	11053	93529	139831	3.151170
2001–3000	1000	19282	158941	280927	3.349864
3001–4000	1000	26482	237073	523987	3.275531
4001–5000	1000	37141	301867	549667	3.650981

The large-scale absolute records after $e = 1000$ occur at

$$1310, 2040, 2079, 2744, 3273, 3414, 3794, 3903, 4718, \\ p(N_e) = 139831, 187597, 276037, 280927, 302299, 327559, 348571, 523987, 549667.$$

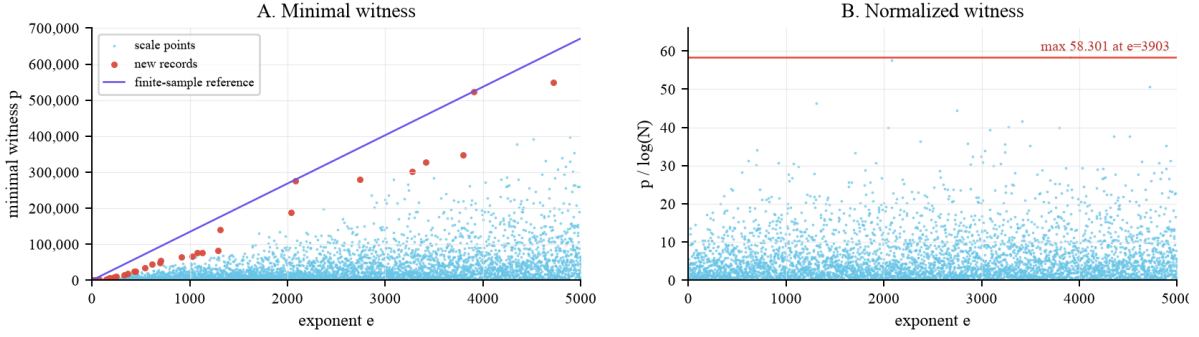


Figure 2: Scale-point minimal witnesses and normalized witness ratios for $N_e = 2 \cdot 10^e$, $e = 2, \dots, 5000$. Red points in panel A mark new absolute records; the blue line is a finite-sample reference, not a bound.

7 A Targeted Sparse-Scale Experiment at $e = 15000$

The scale-point grid $N_e = 2 \cdot 10^e$ is intentionally simple, but it is also sparse in a restrictive way: it samples only one even integer per exponent. After the $e \leq 5000$ table and the later $e = 10000$ offset tests, we ran a more targeted experiment around

$$N = 2 \cdot 10^{15000} + \Delta$$

with small even offsets Δ . The motivation was not that larger e should automatically produce a larger witness. Rather, the normalization changes the threshold. At $e = 10000$, $p = 10^6$ corresponds to

$$p / \log N \approx 43.428140873,$$

whereas at $e = 15000$ it corresponds to

$$p / \log N \approx 28.952384424.$$

The previous offset record at $e = 10000$,

$$p(2 \cdot 10^{10000} - 38602) = 634091,$$

had $p / \log N \approx 27.537393274$. Thus a million-sized witness at $e = 15000$ required only a modest normalized improvement over that local record.

The experiment used a three-stage procedure.

1. A q-free pressure scout scanned offsets

$$-100000 \leq \Delta \leq 100000, \quad \Delta \equiv 0 \pmod{2},$$

with `pLimit` = 10000, small-factor depths 97 and 997, and retained the top 200 offsets.

2. The retained offsets were converted into full GMP input rows and scored by the q-free survivor-road scorer to $p_{\max} = 1,000,000$, using `smallLimit` = 250000. This scorer does not test $q = N - p$ for primality. It records only the admissible modulo-6 lane, the small-prime blockers of q , survivor density, milestone positions such as p at the 1000th survivor, and bucketed road geometry.

3. The exact basket was frozen before witness computation. It mixed the strongest $N \bmod 6 = 4$ road-bucket rows with the strongest $N \bmod 6 = 2$ rank-sum rows from the scorer output. This was a selection rule for a small experiment, not a proof that all high-witness offsets must pass through the same bucket.
4. A small exact basket of ten offsets was then checked by the increasing minimal-witness worker with 18 threads and `GMP primalityReps = 5`.

Table 6 gives the exact results for that basket. One row, $\Delta = -77368$, had no witness below 10^6 . A deeper single-row run to $p_{\max} = 2,000,000$ found the first witness at $p = 1,223,309$.

Table 6: Targeted $e = 15000$ offset basket. Rows are probable-prime computational witnesses under the GMP test used here.

offset Δ	10^6 pass	final $p(N)$	$p/\log N$	pass s	deep s
-77368	no witness	1,223,309	35.417712438	1845.52	2257.19
64214	432,737	432,737	12.528767979	883.538	—
69758	386,537	386,537	11.191167818	823.556	—
64268	381,047	381,047	11.032219228	915.606	—
41828	358,607	358,607	10.382527721	743.347	—
46362	299,191	299,191	8.662292848	695.728	—
89862	190,369	190,369	5.511636471	441.525	—
9138	127,837	127,837	3.701185968	295.304	—
-10134	90,373	90,373	2.616513838	234.865	—
-93048	7,459	7,459	0.215955835	38.746	—

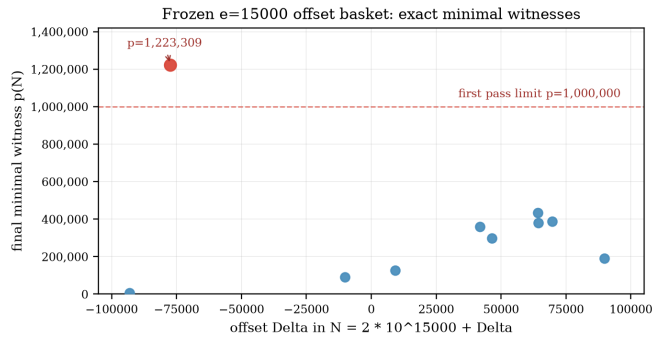


Figure 3: Exact results for the frozen $e = 15000$ offset basket. The highlighted row is the offset that had no witness below 10^6 and was then resolved at $p(N) = 1,223,309$.

The result is encouraging but should be stated carefully. The least-prime Goldbach partition $p(N)$ is a standard object in the literature and in the Oliveira e Silva–Herzog–Pardi data (Ref. 6). The present experiment does not introduce that object, does not verify a consecutive range, and does not prove a predictor. Its narrower contribution is a reproducible sparse-scale measurement: a q-free road-scoring pass selected a small basket that contained a row with $p(N) > 10^6$, and the exact worker then located the witness $p(N) = 1,223,309$.

The negative information is also useful. Several highly ranked rows produced ordinary witnesses, and the $N \bmod 6 = 2$ rank-sum candidates were noisier than the $N \bmod 6 = 4$ road bucket in this run. The current scorer is therefore an enrichment device, not a no-skip rule.

8 Survivor Pressure

The first witness $p(N)$ records where the increasing search stops. To study why the search lasted that long, we also record the failures before the first success. After the modulo-6 lane and the small-prime wall have removed the obvious composites, some candidates remain for which

$$q = N - p$$

is not divisible by the small primes but is still composite. We call the number of such failed candidates before the first witness the survivor pressure of N , for the chosen small-prime bound.

More formally, fix a small-prime bound L . For an admissible prime candidate $p < p(N)$, define the small-prime wall to block p if there is a prime r , $5 \leq r \leq L$, such that

$$r \mid (N - p).$$

The survivor pressure is then

$$S_L(N) = \#\{p < p(N) : p \text{ prime, } p \text{ admissible, } \\ N - p \text{ passes the small-prime wall but is composite}\}.$$

In this note survivor pressure is a post-hoc diagnostic and ranking statistic. It is not an independent formula for predicting $p(N)$, because it is measured only up to an already known first witness.

Table 7: Top scale points by survivor pressure, $201 \leq e \leq 5000$.

e	$p(N_e)$	$p/\log N$	pressure	small-wall blocks	admissible
4718	549667	50.593920	3591	19032	22624
3903	523987	58.300567	3397	18268	21666
4887	397519	35.324263	2648	14143	16792
4514	391873	37.699808	2602	13964	16567
4349	378253	37.770021	2544	13505	16050
4929	354169	31.203946	2374	12749	15124
3794	348571	39.897325	2358	12542	14901
4641	331663	31.034275	2293	11957	14251
4811	340381	30.724660	2290	12287	14578
4885	329617	29.302359	2267	11901	14169

Across the 4800 scale-point rows $e = 201, \dots, 5000$, the Pearson correlation between $p(N_e)$ and survivor pressure is 0.997643. This high value is expected because pressure is measured up to the already-known first witness. The correlation between $p(N_e)/\log N_e$ and survivor pressure is lower, 0.851063, because the normalized quantity also depends on the size of N_e .

The empirical use of survivor pressure is therefore modest but real. It describes the path that the minimal-witness search had to take, and it gives a way to rank already computed scale points for local follow-up.

9 Q-free Wall-Spectrum Triage

Survivor pressure is post-hoc because it is measured only up to a known first witness. A more useful question is whether a related signal can be computed before the exact witness search. The wall-spectrum experiments test this question.

For an even N , and for early admissible prime candidates p , define a candidate to be blocked at wall depth L if $N - p$ has a prime divisor at most L . This uses only divisibility by small primes. It does not test whether $N - p$ is prime and it does not use the exact value of $p(N)$. Across a local window, the resulting small-divisor pressure forms a wall-spectrum: a sequence of local pressure highs and lows across several wall depths.

The current best frozen policy is called K3 adaptive wall persistence. In plain terms, it:

1. scores several wall-depth layers;
2. finds local q-free pressure maxima in each layer;
3. keeps centers that persist across layers;
4. opens small adaptive neighborhoods around the strongest centers;
5. uses exact minimal-witness labels only afterward, for evaluation.

For reproducibility, Table 8 lists the frozen K3 parameters used in the frontier below. The feature layers use equal p -candidate and wall limits at each depth.

Table 8: Frozen K3 adaptive wall-persistence parameters.

Parameter	value
feature layers	5000, 8000, 9000, 10000, 12000
window radius	± 2000 rows
labels	minimal-witness labels used only for evaluation
label limits	<code>pLimit</code> = 50000, <code>smallLimit</code> = 50000
layer local suppression	<code>layerSuppress</code> = 20
layer centers	<code>layerTopK</code> = 3
persistence vote radius	<code>voteR</code> = 20
final center suppression	<code>finalSuppress</code> = 100
final centers	<code>finalTopK</code> = 10
base reveal radius	<code>baseR</code> = 20
adaptive expansion	<code>expandTop</code> = 4, <code>expandR</code> = 60

The benchmark uses known large-witness rows from the Oliveira e Silva–Herzog–Pardi least-prime data (Ref. 6). One benchmark group consists of a known least-prime row and the contiguous ± 2000 -row local window around it. The feature stage receives the even values in that local window, computes only q-free small-divisor features, and opens a subset of rows. A hit means that the opened subset contains the known least-prime row. Exact minimal witnesses are used only afterward as labels for evaluation; they are not available to the feature stage.

Table 9 gives the current comparable frontier. The nine fresh holdout blocks cover least-prime ranks 51–500, grouped into fifty-row windows. Each policy in the table is measured on the same 450 benchmark groups. The “dense” baselines use single-layer dense local maxima with fixed or adaptive reveal radii; K3 uses persistence across the five feature layers in Table 8.

The positive result is that K3 adaptive covers substantially more target rows than dense wide while opening less than half as many rows. The negative result is equally important: K3 adaptive still misses 37/450 target rows. The tested rule is therefore a triage and enrichment method, not a formula and not a no-skip localizer.

Table 9: Q-free wall-spectrum holdout frontier on least-prime local windows.

Policy	holdouts	hits	hit rate	mean opened rows
K3 adaptive	9	413/450	91.78%	368.30
dense wide	9	346/450	76.89%	807.96
dense adaptive	9	298/450	66.22%	347.36
dense strict	9	287/450	63.78%	206.89

Two variants were also checked because earlier inspected blocks made them look promising. The fresh candidate-validation block was least-prime ranks 451–500. It did not promote either variant over K3, as shown in Table 10.

Table 10: Fresh candidate validation on least-prime ranks 451–500.

Policy	hits	mean opened rows	interpretation
K3 adaptive	48/50	368.32	current baseline
MassR100 priority	47/50	365.20	cheaper, but loses one hit
confidence tier	48/50	442.04	ties K3, but costs more

The two K3 misses in the fresh candidate-validation block are informative. In both cases the true target was near a selected wall center, but the opened radius was too small. This supports the same interpretation: the wall-spectrum signal often lands near hard rows, but the tested reveal geometry is not safe enough to discard all unopened rows.

10 Q-line Interval Coverage

The q-line calculation is a different computation. It does not compute $p(N)$. It tries to prove a weaker finite statement: for every even N in a stated finite block, find at least one decomposition $N = p + q$ under the primality tests used by the worker.

The idea is classical. Shen used a digital computer to check Goldbach up to 33,000,000 (Ref. 1), and Sinisalo later checked up to $4 \cdot 10^{11}$ (Ref. 2). Granville, te Riele, and van de Lune studied checking Goldbach on a vector computer (Ref. 3). Deshouillers, te Riele, and Saouter later described interval algorithms in terms of finding sets of primes P and Q such that $P + Q$ covers all even values in an interval (Ref. 4). Their Cray implementation used bit arrays and shifted copies of prime arrays. The present q-line worker uses the same general reversal: choose many prime q ’s near the interval, then use a prime bitset for the smaller moving part p .

For one block, let N_0 be a chosen anchor near the block. For an odd value a , define

$$q_a = N_0 - a.$$

If q_a is prime, then for each even N in the block the corresponding candidate is

$$p_a(N) = N - q_a = a + (N - N_0).$$

The q-line covers exactly the rows for which $p_a(N)$ is also prime. One prime q_a can therefore cover many rows.

The worker builds many such lines. Candidate q_a ’s are first filtered by small primes up to `qFilterLimit`, then tested with GMP. The moving p -side is represented by an ordinary prime

bitset over the needed integer range. The block coverage bitset is updated by OR-ing the shifted prime slices from all accepted q-lines.

Rows not covered by q-lines are not discarded. Each uncovered row is sent to a direct fallback search. A finite block is accepted only when

$$\text{qlineRows} + \text{fallbackVerified} = \text{rows}, \quad \text{unresolved} = 0.$$

This no-skip condition is the important safety rule. A q-line miss is only a miss by the current q-line geometry; it is not evidence against Goldbach and not automatically a minimal-witness peak.

11 The $e19$ -forward Q-line Campaign

The completed q-line campaign reported here starts at

$$B = 2 \cdot 10^{19}$$

and scans consecutive even values

$$N = B + \text{offset}.$$

The offsets range from 2 to 2,004,000,000,000. Equivalently, the checked even interval is

$$2 \cdot 10^{19} + 2 \leq N \leq 2 \cdot 10^{19} + 2,004,000,000,000.$$

The interval contains 1,002,000,000,000 even values, split into 1002 blocks of 1,000,000,000 rows.

The campaign parameters are shown in Table 11.

Table 11: Parameters for the $e19$ -forward q-line campaign.

Parameter	value
rows per block	1,000,000,000
blocks	1002
aLimit	50,000
pLimit for direct fallback	250,000
qFilterLimit	250,000
smallLimit for fallback	250,000
GMP repetitions	25
block-parallel jobs	18

The aggregate result is shown in Table 12.

The result should be phrased narrowly:

Every tested even N in this finite interval had at least one
Goldbach decomposition under the GMP probable-prime tests used here.

This is not a statement about untested integers. It is not a minimal-witness profile over the interval. It is also not a formal certificate unless the accepted large primes are later independently certified.

Table 12: Aggregate result of the $e19$ -forward q-line campaign.

Quantity	value
even values checked	1,002,000,000,000
rows covered by q-lines	1,001,999,948,917
q-line coverage share	99.999994902%
direct fallback rows	51,083
direct fallback rows verified	51,083
unresolved rows	0
accepted q-lines, summed by block	1,127,333
GMP tests for q-line candidates	3,388,627
GMP tests in direct fallback	167,760
total q-side GMP tests	3,556,387
average fallback rows per billion	50.981038
minimum fallback rows in a block	4
maximum fallback rows in a block	207

12 Contribution and Limits

The q-line idea itself is not new; the relevant prior art is explicitly cited above and in the references. The appropriate claim is more limited: the repository provides a compact modern implementation, a reproducible data snapshot, and engineering measurements for a large interval above $2 \cdot 10^{19}$ under a clearly stated probable-prime convention.

The same caution applies to minimal Goldbach witnesses. The least-prime partition $p(N)$, first-occurrence tables $S(p)$, and record-holders for minimal Goldbach partitions are prior work, not inventions of this project. The Oliveira e Silva project page explicitly defines the minimal Goldbach partition and reports top least-prime record data below $4 \cdot 10^{18}$ (Ref. 6). The targeted $e = 15000$ result in this note is different in scope: it is a sparse, very large decimal-scale point, selected by q-free triage and checked under GMP probable-prime tests. It should be read as an experimental measurement and a test of the triage workflow.

The most useful observations are practical. Minimal-witness data, wall-spectrum triage, and q-line coverage answer different questions. The first measures $p(N)$. The second tries to choose promising neighborhoods before the exact witness search. The third finds any valid decomposition in a finite block. On the scale-point grid $2 \cdot 10^e$, $p(N_e)$ behaves like a noisy spike process rather than a monotone formula. Survivor pressure is a diagnostic of an already computed row, not a proven predictor. The wall-spectrum benchmark shows a reproducible q-free enrichment signal: K3 adaptive covers more known benchmark target rows than dense wide at lower opened-row cost. It also shows the boundary of the present method: the rule still misses target rows, and the first fresh MassR100/tier candidate validation did not improve the K3 frontier.

In the q-line campaign, almost all rows were covered by shifted prime bitsets, and the tiny residue was handled by direct fallback. The fallback layer is the implementation detail that keeps the method no-skip for interval coverage. The resulting finite coverage is reproducible, but the present standard is probable-prime computation; independent primality certificates have not yet been produced for these rows.

In the sparse offset campaign, the q-free survivor-road scorer found a basket containing

$$N = 2 \cdot 10^{15000} - 77368, \quad p(N) = 1,223,309.$$

This is the strongest single minimal-witness row currently reported by this project. The restrained conclusion is that the road-score triage can enrich a small exact basket for unusually delayed witnesses. The equally important limitation is that the same basket contained several ordinary rows; the method is not a formula for locating all large witnesses.

The present data therefore justify an open repository and a computational note. They do not justify claiming a new proof of Goldbach, a no-skip formula for minimal witnesses, a new discovery of q-line coverage, or a certified extension of the known complete verification.

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